MH4110 Partial Differential Equations

Chapter 4 - Boundary problems

Outline

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- Boundary Value Problems
- Separation of Variables: Homogeneous Dirichlet conditions
  - *Wave equations*
  - *Heat equations*
- Separation of Variables: Homogeneous Neumann conditions
  - *Wave equations*
  - *Heat equations*

Initial Value/Cauchy Problems

Wave equation on the whole line:

$$u_{tt} = c^2 u_{xx}, \quad x \in \mathbb{R} = (-\infty, \infty), \quad t > 0,$$
$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad x \in \mathbb{R},$$

where ϕ, ψ are any given functions.

Solution formula–d'Alembert formula:

$$u(x,t) = \frac{1}{2}\big[\phi(x+ct) + \phi(x-ct)\big] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds.$$

Imagine the picture of the solution with $\psi = 0$.

Initial Value/Cauchy Problems (Cont'd)

Heat equation on the whole line

$$u_t = ku_{xx}, \quad x \in \mathbb{R} = (-\infty, \infty), \quad t > 0,$$
$$u(x, 0) = \phi(x), \quad x \in \mathbb{R},$$

where ϕ is a given function.

Solution formula:

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\Big(- \frac{(x-y)^2}{4kt} \Big) \phi(y) dy.$$

Let $S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}$. Then we can write the solution as a convolution:

$$u(x,t) = S(\cdot, t) * \phi.$$

Initial Boundary Value Problems

Wave equation on the half line with Dirichlet boundary conditions

$$
\begin{aligned}
&v_{tt} - c^2 v_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&v(x,0) = \phi(x), \quad v_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0), \\
&v(0,t) = 0 \quad \text{(boundary condition at } x = 0),
\end{aligned}
$$

where we assume that $\phi(0) = 0$ (consistent condition).

Solution formula:

$$
v(x,t) = \begin{cases} \dfrac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\ \dfrac{1}{2}\left[\phi(x+ct) - \phi(ct-x)\right] + \dfrac{1}{2c}\displaystyle\int_{ct-x}^{ct+x} \psi(s)ds, & 0 < x < ct. \end{cases}
$$

Initial Boundary Value Problems (Cont'd)

Wave equation on the half line with Neumann boundary conditions

$$w_{tt} - c^2 w_{xx} = 0, \quad 0 < x < \infty, \quad t > 0,$$
$$w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0\text{)},$$
$$w_x(0,t) = 0 \quad \text{(boundary condition at } x = 0\text{)}.$$

Solution formula:

$$w(x,t) = \begin{cases} \dfrac{\phi(x+ct) + \phi(x-ct)}{2} + \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\[2ex] \dfrac{\phi(x+ct) + \phi(ct-x)}{2} + \dfrac{1}{2c}\left(\displaystyle\int_0^{ct-x} + \int_0^{x+ct}\right)\psi(s)ds, & 0 < x < ct. \end{cases}$$

Initial Boundary Value Problems (Cont'd)

Heat equation on the half line with Dirichlet boundary conditions

$$
\begin{aligned}
&v_t - kv_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&v(x,0) = \phi(x), \quad \text{(initial condition at } t = 0), \\
&v(0,t) = 0 \quad \text{(boundary condition at } x = 0).
\end{aligned}
$$

Solution formula:

$$
v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) - \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.
$$

Initial Boundary Value Problems (Cont'd)

Heat equation on the half line with Neumann boundary conditions

$$
\begin{aligned}
&w_t - kw_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\
&w(x,0) = \phi(x), \quad \text{(initial condition at } t = 0), \\
&w_x(0,t) = 0 \quad \text{(boundary condition at } x = 0).
\end{aligned}
$$

Solution formula:

$$
w(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) + \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.
$$

§4.1 Boundary value problems

The absence of boundaries is a big simplification, but most of physical problems have fixed boundaries. Therefore, it is necessary to consider boundary value problems.

Wave equation with homogeneous Dirichlet boundary conditions

The wave equation

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \tag{1}$$

is subjected to the homogeneous boundary conditions

$$u(0,t) = u(l,t) = 0, \quad t > 0, \tag{2}$$

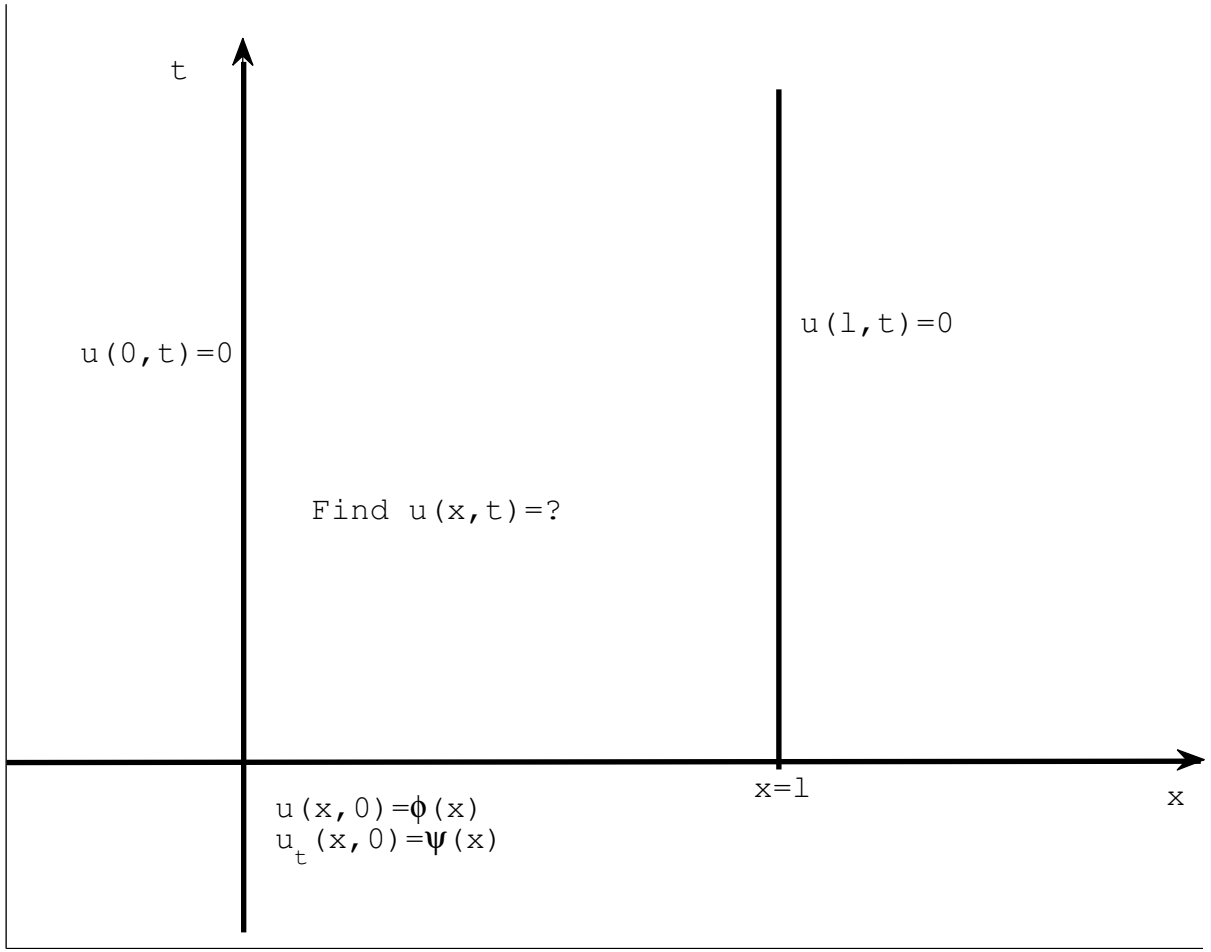
and the initial condition

$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \tag{3}$$

where $c, l > 0$ are given constants, and ϕ, ψ are given functions.

§4.1 Boundary value problems (Cont'd)

Wave equations with Dirichlet boundary conditions on a finite interval



§4.1 Separation of Variables: The Dirichlet condition

Separation of Variables

Look for a **separated solution** of (39) of the form

$$u(x,t) = X(x)T(t), \tag{4}$$

which should satisfy the initial and boundary conditions (2)-(3). This form will be substituted into (39) and result in ODEs for X and T, respectively. The method of splitting x and t dependency as in (4) is called **separation of variables**.

We shall see that the applicability of this method essentially depends on the type of equations and solution domains (and type of boundary conditions sometimes).

§4.1 Separation of Variables–Method

Step I: Plugging (4) into (39) leads to

$$X(x)T''(t) = c^2 X''(x)T(t). \tag{5}$$

Dividing both sides of the identity by $-c^2 X(x)T(t)$ to separate the variables, we get

$$-\frac{X''(x)}{X(x)} = -\frac{T''(t)}{c^2 T(t)} := \lambda. \tag{6}$$

Two observations are in order.

(i) λ must be a **constant**. As x and t are two independent variables, we find that

$$\frac{\partial \lambda}{\partial t} = \frac{\partial}{\partial t}\left(-\frac{X''(x)}{X(x)}\right) = 0; \quad \frac{\partial \lambda}{\partial x} = \frac{\partial}{\partial x}\left(-\frac{T''(t)}{c^2 T(t)}\right) = 0.$$

Thus, λ is a constant.

(ii) We will shortly see that the boundary conditions force $\lambda > 0$ to be positive, so we put minus signs in (6).

§4.1 Separation of Variables–Method (Cont'd)

So let $\lambda = \beta^2$ with $\beta > 0$. Then we arrive at a pair of **separate** ODEs:

$$X''(x) + \beta^2 X(x) = 0; \quad T''(t) + c^2\beta^2 T(t) = 0. \tag{7}$$

Step II: Solve the ODEs (7) and find the value of λ (i.e., β) so that the solution satisfies the boundary conditions (2).

We started with the first ODE, and find that it has the **general solution**:

$$X(x) = C\cos\beta x + D\sin\beta x, \tag{8}$$

where C and D are two arbitrary constants.

To meet the boundary conditions (2), we require that

$$u(0,t) = X(0)T(t) = 0 = u(l,t) = X(l)T(t) \Rightarrow X(0) = X(l) = 0. \tag{9}$$

Note that $T(t) \equiv 0$ would result in the trivial solution $u(x,t) = 0$.

§4.1 Separation of Variables–Method (Cont'd)

Then, we find from (8) that

$$X(0) = C = 0, \quad X(l) = D \sin \beta l = 0. \tag{10}$$

The solution with $D = 0$ will again lead to the trivial zero solution, so we consider the case when $\sin \beta l = 0$, implying that

$$\sin \beta l = 0 \quad \Rightarrow \quad \beta l = n\pi \quad \Rightarrow \quad \beta = \frac{n\pi}{l}, \tag{11}$$

where n could be any integer at this point.

Step III: Formulate the solution.
We find from the above that

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{l}. \tag{12}$$

For each n, the pair (λ_n, X_n) is a solution of the **eigen-problem**:

$$X'' = -\lambda X, \quad X(0) = X(l) = 0. \tag{13}$$

§4.1 Separation of Variables–Method (Cont'd)

Now, we turn to the second equation of (7) with β given by (11). We find that the general solution is

$$T_n(t) = A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l}, \tag{14}$$

for each $n = 1, 2, 3, \cdots$, where A_n and B_n are arbitrary constants.

Therefore, the boundary value problem (39)-(2) (we have not imposed the initial condition (3) yet) have infinite number of separated solutions:

$$u_n(x,t) = X_n(x)T_n(t) = \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l}\right) \sin \frac{n\pi x}{l}, \tag{15}$$

for all $n = 1, 2, 3, \cdots$.

§4.1 Separation of Variables–Method (Cont'd)

Based on the **principle of superposition**, the series

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l} \tag{16}$$

is also a solution of (39)-(2).

Step IV: Use the initial condition (3) to determine A_n and B_n

$$\phi(x) = u(x,0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l},$$

$$\psi(x) = u_t(x,0) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}.$$

In Chapter 5, we shall use the Fourier series of ϕ and ψ to find A_n and B_n.

Question: Why $\lambda > 0$?

§4.1 Separation of Variables–Method (Cont'd)

Now, we come to this issue by showing that $\lambda \leq 0$ does not provide any meaningful solutions.

- **Case I: $\lambda = 0$.** In this case, the first equation of (6) becomes $X'' = 0$, so $X(x) = ax + b$. To meet the boundary conditions (9), we find $a = b = 0$ so $X(x) = 0$. The solution of the form (4) is trivial, so we reject $\lambda = 0$.

- **Case II: $\lambda < 0$.** In this case, the first equation of (6) becomes

$$X''(x) + \lambda X(x) = 0 \quad \Rightarrow \quad X(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}.$$

  By the boundary conditions $X(0) = X(l) = 0$, we find that

$$c_1 + c_2 = 0, \quad c_1 e^{-\sqrt{-\lambda}l} + c_2 e^{\sqrt{-\lambda}l} = 0.$$

  This 2-by-2 linear system with unknowns c_1, c_2 has only zero solution, i.e., $c_1 = c_2 = 0$. Thus, once again, we obtain trivial solution $u(x,t) = 0$, so we reject $\lambda < 0$ either.

§4.1 Separation of Variables–Method (Cont'd)

Theorem 1

The wave equation with homogeneous Dirichlet boundary conditions (39)-(3) has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}, \tag{17}$$

where A_n and B_n are to be determined by the initial condition (3).

Remark: The coefficients of t inside the series (17), $n\pi c/l$, are called the frequencies. The smallest frequency, $\pi c/l$, is the fundamental note, while the double, triple, and so on of the fundamental note are the overtones. Notice that by shortening the length l of the vibrating portion of the string with a finger, a violinist produces notes of higher frequency.

§4.1 Separation of Variables–Example

Example A (not in the textbook)

Use the method of separation of variables to obtain a solution of the wave equation:

$$u_{tt} = u_{xx}, \quad x \in (0, 1),$$

subject to the conditions:

$$\begin{aligned} &u(0, t) = u(1, t) = 0, \\ &u(x, 0) = \sin \pi x + \sin(2\pi x), \quad u_t(x, 0) = 0. \end{aligned}$$

§4.1 Separation of Variables–Example (Cont'd)

§4.1 Separation of variables – Method

Heat equation with homogeneous Dirichlet boundary conditions

Consider the boundary value problem

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \;\; 0 < t < \infty, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{18}$$

where k, l are positive constants and ϕ is a given function.

Separation of Variables

(i) Look for solution of the form: $u(x,t) = X(x)T(t)$. In this case, we get

$$-\frac{X''}{X} = -\frac{T'}{kT} := \lambda.$$

Once again, λ must be a constant (why?).

§4.1 Separation of Variables–Method

(ii) We have the same eigenproblem as

$$X'' = -\lambda X, \quad X(0) = X(l) = 0, \tag{19}$$

which admits the eigen-pairs

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots. \tag{20}$$

(iii) Solve $T' + \lambda k T = 0$, and obtain

$$T_n(t) = A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big).$$

(iv) The solution is

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin\frac{n\pi x}{l}.$$

In view of $u(x,0) = \phi(x)$, with A_n to be determined we deduce that

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l}.$$

Theorem 2

The heat equation (18) has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin\frac{n\pi x}{l}, \tag{21}$$

provided that ϕ has the expansion

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l}. \tag{22}$$

In Chapter 5, we shall show how to find A_n from ϕ.

§4.1 Separation of Variables–Example

Example B (not in the textbook)

Solve the following Dirichlet problem for the heat equation by separation of variables.

$$
\begin{aligned}
&u_t = u_{xx}, \quad x \in (0, \pi/2), \\
&u(0,t) = u(\pi/2, t) = 0, \\
&u(x,0) = 2\sin 6x.
\end{aligned}
$$

§4.1 Separation of Variables–Example

Example C (not in the textbook)

Consider the damped wave equation

$$\begin{aligned} &u_{tt} + u_t = u_{xx}, \quad x \in (0,1),\ t > 0, \\ &u(0,t) = u(1,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad x \in (0,1). \end{aligned} \tag{23}$$

§4.1 Separation of Variables–Example (Cont'd)

§4.1 A short summary

- For the Dirichlet problems for the wave and heat equations on a finite interval, we solved them with the method of separation of variables. That is, we looked for the solution in the form of an infinite linear combination of separated solutions. This lead to infinite series, which solve the appropriate initial-boundary value problems, as long as the initial data can be expanded in corresponding series, which in the case of Dirichlet conditions are the Fourier sine series.
- It is essential to solve the eigenproblem for the operator $-\frac{d^2}{dx^2}$ in the space of functions satisfying the boundary conditions:

$$-X''(x) = \lambda X(x), \quad X(0) = X(l) = 0, \tag{24}$$

  whose eigenvalues and eigenfunctions are:

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots .$$

- However, unlike the operators in linear algebra, which have finitely many eigenvalues, in our case we have an infinite number of eigenvalues. This is due to the fact that the space of functions is infinite dimensional.

§4.2 Separation of variables: The Neumann condition

Wave equation with homogeneous Neumann boundary conditions

Solve the wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0. \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l. \end{aligned} \tag{25}$$

Substituting the separated solution $u(x,t) = X(x)T(t)$ into the wave Neumann problem gives the same equations for X and T as in the Dirichlet case,

$$X'' = -\lambda X, \quad -T'' = c^2 \lambda T. \tag{26}$$

However, the boundary conditions now imply

$$X'(0)T(t) = X'(l)T(t) = 0, \quad \forall t \to X'(0) = X'(l) = 0.$$

Once again, $\lambda \geq 0$ gives meaningful solutions (why?).

§4.2 Neumann Boundary Conditions (Cont'd)

Assume that $\lambda = \beta^2 \geq 0$. In this case, the general solution is

$$X(x) = C\cos\beta x + D\sin\beta x$$

and

$$X'(x) = -C\beta\sin\beta x + D\beta\cos\beta x.$$

Therefore, $X'(0) = 0$ implies $D = 0$, and $X'(l) = 0$ implies $C\sin\beta l = 0$. Since we do not want $C = 0$, we have to take $\beta l = n\pi$ for any integer n. This leads to the eigen-pairs of (26):

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \cos\frac{n\pi x}{l}, \quad n = 0, 1, 2, \cdots, \tag{27}$$

as opposed to the Dirichlet boundary condition case. Also notice that $\lambda = 0$ is also an eigenvalue of (26) with the corresponding eigenfunction 1.

§4.2 Neumann Boundary Conditions (Cont'd)

For $T(t)$,

$$T'' = -c^2 \lambda T, \quad \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad n = 0, 1, 2, \cdots .$$

When $n = 0$, $T_0'' = 0$, so

$$T_0(t) = \frac{A_0}{2} + \frac{B_0}{2} t,$$

where A_0 and B_0 are constants to be determined by the initial condition. Here, the factor $1/2$ is conventional for the formulation of Fourier series (see Chapter 5).

When $n \geq 1$, the expressions of $T_n(t)$ are given by

$$T_n(t) = A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l}. \tag{28}$$

§4.2 Neumann Boundary Conditions (Cont'd)

Theorem 3

The wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \end{aligned} \tag{29}$$

has the following series solution:

$$u(x,t) = \frac{A_0}{2} + \frac{B_0}{2}t + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \cos \frac{n\pi x}{l}, \tag{30}$$

as long as the initial data can be expanded into cosine Fourier series

$$\phi(x) = \frac{1}{2}A_0 + \sum_n A_n \cos \frac{n\pi x}{l}, \quad \psi(x) = \frac{1}{2}B_0 + \sum_n \frac{n\pi c}{l} B_n \cos \frac{n\pi x}{l}. \tag{31}$$

§4.1 Separation of Variables–Example

Example D (not in the textbook)

Solve the following Neumann problem for the wave equation by separation of variables.

$$
\begin{aligned}
&u_{tt} - 4u_{xx} = 0, \quad x \in (0, \pi), \\
&u_x(0, t) = u_x(\pi, t) = 0, \\
&u(x, 0) = 3\cos x, \quad u_t(x, 0) = 1 - \cos 4x.
\end{aligned}
$$

§4.2 Neumann Boundary Conditions– Heat equation

Theorem 4

The heat equation

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{32}$$

has the series solution:

$$u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \cos\frac{n\pi x}{l}, \tag{33}$$

provided that ϕ has the cosine Fourier expansion

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}. \tag{34}$$

In Chapter 5, we shall show how to find A_n from ϕ.

§4.2 Mixed Boundary Conditions– Example

Sometimes one needs to consider problems with mixed Dirichlet-Neumann boundary conditions, i.e. Dirichlet conditions at one end of the finite interval, and Neumann conditions at the other.

Example E

Consider the heat equation with "mixed" boundary conditions:

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \ t > 0, \\ &u(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l. \end{aligned}$$

Derive the solution formula.

Solution: Assume that $u(x,t) = X(x)T(t)$. It is essential to solve the eigenproblem:

$$X'' + \lambda X = 0, \quad X(0) = X'(l) = 0.$$

§4.2 Mixed Boundary Conditions– Example (Cont'd)

- For $\lambda > 0$, we find that

$$X(x) = C\cos\sqrt{\lambda}x + D\sin\sqrt{\lambda}x.$$

Therefore, $X(0)$ implies $C = 0$, and $X'(l) = D\sqrt{\lambda}\cos\sqrt{\lambda}l = 0$. Since we do not want D to be zero, we have to take

$$\lambda_n = \left(n + \frac{1}{2}\right)^2 \frac{\pi^2}{l^2}, \quad X_n(x) = \sin\left(\left(n + \frac{1}{2}\right)\frac{\pi x}{l}\right),$$

for $n = 0, 1, \cdots$.

- For $\lambda = 0$, we see that $X'' = 0$ and $X(x) = ax + b$. Hence, $X(0) = b = 0$ and $X'(l) = a = 0$. This gives trivial solution $u = 0$, which should be rejected.

- For $\lambda < 0$, we find

$$X(x) = Ce^{-\sqrt{-\lambda}x} + De^{\sqrt{-\lambda}x}.$$

By requiring $X(0) = X'(l) = 0$, we can show that $C = D = 0$. Therefore, $\lambda < 0$ is not possible.

§4.3 Mixed Boundary Conditions– Example (Cont'd)

The equation for $T(t)$ is still the same. Therefore, we obtain the solution formula:

$$u(x,t) = \sum_{n=0}^{\infty} A_n \exp\Big(-\Big(n+\frac{1}{2}\Big)^2 \frac{\pi^2}{l^2} kt \Big) \sin\Big(\Big(n+\frac{1}{2}\Big)\frac{\pi x}{l} \Big), \tag{35}$$

where again A_n's are determined by ϕ.

If the boundary condition is alternatively as $u_x(0,t) = u(l,t) = 0, \quad t > 0$, then the solution formula is

$$u(x,t) = \sum_{n=0}^{\infty} A_n \exp\Big(-\Big(n+\frac{1}{2}\Big)^2 \frac{\pi^2}{l^2} kt \Big) \cos\Big(\Big(n+\frac{1}{2}\Big)\frac{\pi x}{l} \Big), \tag{36}$$

A short summary

- Similar to the case of the Dirichlet problems for heat and wave equations, the method of separation of variables applied to the Neumann problems on a finite interval leads to an eigenvalue problem for the $X(x)$ factor of the separated solution.
- In this case, however, we discovered a new eigenvalue $\lambda = 0$ in addition to the eigenvalues found for the Dirichlet problems.
- The general solutions of the Neumann problems for wave and heat equations can be written in series forms, as (infinite) linear combinations of all separated solutions, as long as the initial data can be expanded in cosine Fourier series.

A short summary

We have seen that separation of variables essentially depends on solving the eigen-problem $X'' + \lambda X = 0$ with different types of boundary conditions. Previously, we just discussed the Dirichlet, Neumann and special "mixed" boundary conditions.

Indeed, the method could work for

- more general "mixed" boundary conditions (see Chapter 5);
- inhomogeneous boundary conditions and source terms (see Chapter 5);
- other types of equations, e.g., the Laplace's equation (we will discuss this in Chapter 6, if time allows):

$$\Delta u = u_{xx} + u_{yy} = 0;$$

- other types of separable domains, other than $(0, l) \times (0, \infty)$, e.g., ball and disk, but they are usually much more complicated.

Boundary value problems

Exercise: Find the solution for the wave equation

$$u_{tt} = c^2 u_{xx}, \quad a < x < b, \quad t > 0,$$

subjected to the homogeneous boundary conditions

$$u(a,t) = u(b,t) = 0, \quad t > 0,$$

and the initial condition

$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad a < x < b,$$

where $c > 0$ and a, b are given constants, and ϕ, ψ are given functions.